

Python Programming Notes - Questions and Answers

This PDF contains important Python programming concepts in Question and Answer format. These notes are useful for students, interviews, viva preparation, and beginners learning Python.

Q: 1. What is Python?

Answer: Python is a high-level, interpreted, and object-oriented programming language. It is simple, readable, and widely used in web development, AI, machine learning, and automation.

Q: 2. What are the features of Python?

Answer: Python features include simplicity, portability, interpreted execution, dynamic typing, object-oriented programming, and a large standard library.

Q: 3. What is an interpreted language?

Answer: An interpreted language executes code line by line using an interpreter instead of compiling the entire program at once.

Q: 4. What are variables in Python?

Answer: Variables are containers used to store data values. Python does not require explicit declaration of variable types.

Q: 5. What are Python data types?

Answer: Python data types include int, float, str, list, tuple, set, dictionary, and boolean.

Q: 6. What is type casting in Python?

Answer: Type casting converts one data type into another using functions like `int()`, `float()`, and `str()`.

Q: 7. What are operators in Python?

Answer: Operators perform operations on variables and values. Types include arithmetic, relational, logical, assignment, and bitwise operators.

Q: 8. Explain if-else statements in Python.

Answer: if-else statements are used for decision making. They execute different blocks of code based on conditions.

Q: 9. What are loops in Python?

Answer: Loops repeatedly execute code blocks. Python supports for loops and while loops.

Q: 10. What is a list in Python?

Answer: A list is an ordered and mutable collection used to store multiple values.

Q: 11. What is a tuple in Python?

Answer: A tuple is an ordered and immutable collection of elements.

Q: 12. What is a dictionary in Python?

Answer: A dictionary stores data in key-value pairs and is mutable.

Q: 13. What is a set in Python?

Answer: A set is an unordered collection of unique elements.

Q: 14. What is a function in Python?

Answer: A function is a reusable block of code used to perform a specific task.

Q: 15. What is recursion?

Answer: Recursion is a process where a function calls itself repeatedly until a condition is met.

Q: 16. What is Object-Oriented Programming in Python?

Answer: Object-Oriented Programming (OOP) is a programming approach based on classes and objects.

Q: 17. What is a class in Python?

Answer: A class is a blueprint for creating objects containing variables and methods.

Q: 18. What is an object in Python?

Answer: An object is an instance of a class.

Q: 19. What is inheritance in Python?

Answer: Inheritance allows one class to inherit properties and methods from another class.

Q: 20. What is polymorphism in Python?

Answer: Polymorphism allows methods or objects to take multiple forms.

Q: 21. What is encapsulation?

Answer: Encapsulation binds data and methods together and restricts direct access.

Q: 22. What is exception handling in Python?

Answer: Exception handling manages runtime errors using try, except, finally, and raise blocks.

Q: 23. What are modules in Python?

Answer: Modules are Python files containing functions and variables that can be reused.

Q: 24. What is NumPy?

Answer: NumPy is a Python library used for numerical computations and array operations.

Q: 25. What is Pandas?

Answer: Pandas is a Python library used for data analysis and manipulation.

Q: 26. What is Machine Learning in Python?

Answer: Machine Learning is the process of training computers to learn patterns from data. Python provides libraries like scikit-learn and TensorFlow.

Q: 27. What is Python used for?

Answer: Python is used in web development, automation, data science, machine learning, AI, cybersecurity, and software development.